Design FMEA (qualitative) — VBF-T1OA-DFMEA-01

Qualitative version based on simulation signals; severity/occurrence = H/M/L three-level. Complete FMEA (process/supplier failures) is N/A beyond pure-simulation boundary.

Failure mode	Failure cause	Simulation signal	Severity	Occurrence	Mitigation in design
Negative electrode lithium plating (fast charge)	anode surface potential < 0 V at 4C	`anode_potential_v` min	H	L (measured +0.0341 V margin)	SiOx anode OCP + nanostructured anode (0.8 µm) + high-transport electrolyte (σ 3.0, t■ 0.6)
Thermal runaway risk (temperature exceedance)	high-rate heat generation vs cooling	`T_max_K` 324.55 K vs 333.15 K	H	L (8.6 K margin)	liquid cooling h = 80 $W/m^2/K$
Overcharge thermal runaway	overcharge to 4.7 V drives side reactions	`triggered` = false	H	L	high-voltage-stable design; verified via run-tr
Electrolyte oxidative decomposition (voltage window)	electrolyte oxidation above cutoff	HOMO/IE-EA not computed (real_compute=false)	M	M (not quantified)	advanced electrolyte formulation (estimate)
Insufficient capacity	cathode/anode capacity balance	1C `capacity_ah` 5.057 vs 5.0	L	L	N/P balance per parameter set

Conclusion

Highest-risk items (plating, thermal runaway, overcharge) are mitigated in the design and verified by simulation (all pass). The electrolyte oxidative-decomposition risk is not first-principles-quantified (true DFT skipped, real_compute=false) and is the primary residual uncertainty; the advanced electrolyte transport parameters are marked as estimates. Complete process/supplier FMEA is N/A (beyond pure-simulation boundary).